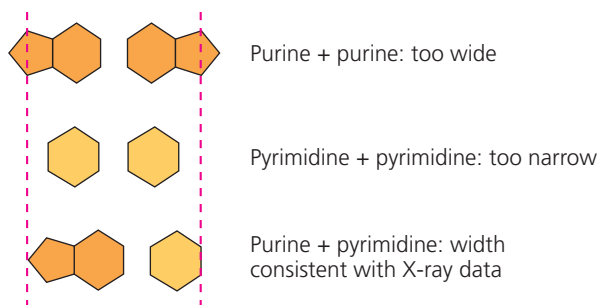


the sugar-phosphate backbones, and the rungs represent pairs of nitrogenous bases. Now imagine twisting the ladder to form a helix. Franklin's X-ray data indicated that the helix makes one full turn every 3.4 nm along its length. With the bases stacked just 0.34 nm apart, there are ten layers of base pairs, or rungs of the ladder, in each full turn of the helix.

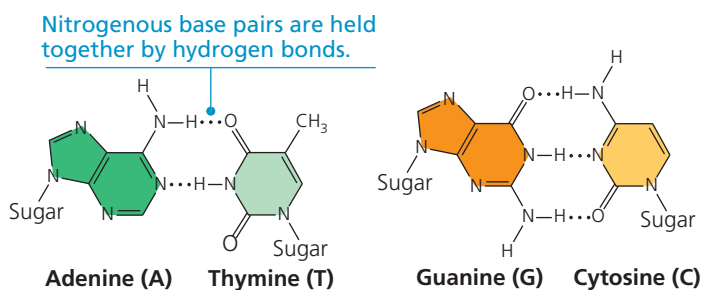
The nitrogenous bases of the double helix are paired in specific combinations: adenine (A) with thymine (T), and guanine (G) with cytosine (C). It was mainly by trial and error that Watson and Crick arrived at this key feature of DNA. At first, Watson imagined that the bases paired like with like—for example, A with A and C with C. But this model did not fit the X-ray data, which suggested that the double helix had a uniform diameter. Why is this requirement inconsistent with like-with-like pairing of bases? Adenine and guanine are purines, nitrogenous bases with two organic rings, while cytosine and thymine are nitrogenous bases called pyrimidines, which have a single ring. Pairing a purine with a pyrimidine is the only combination that results in a uniform diameter for the double helix (**Figure 16.8**).

▼ **Figure 16.8** Possible base pairings in the DNA double helix.



Watson and Crick reasoned that there must be additional specificity of pairing dictated by the structure of the bases. Each base has chemical side groups that can form hydrogen bonds with its appropriate partner: Adenine can form two hydrogen bonds with thymine and only thymine; guanine forms three hydrogen bonds with cytosine and only cytosine. In shorthand, A pairs with T, and G pairs with C (**Figure 16.9**).

▼ **Figure 16.9** Base pairing in DNA.



The Watson-Crick model took into account Chargaff's ratios and ultimately explained them. Wherever one strand of a DNA molecule has an A, the partner strand has a T. Similarly, a G in one strand is always paired with a C in the complementary strand. Therefore, in the DNA of any organism, the amount of adenine equals the amount of thymine, and the amount

of guanine equals the amount of cytosine. (Modern DNA-sequencing techniques have confirmed that the amounts are exactly equal.) Although the base-pairing rules dictate the combinations of nitrogenous bases that form the “rungs” of the double helix, they do not restrict the sequence of nucleotides *along* each DNA strand. The linear sequence of the four bases can be varied in countless ways, and each gene has a unique base sequence.

In April 1953, Watson and Crick surprised the scientific world with a succinct, one-page paper that reported their molecular model for DNA: the double helix, which has since become an icon of molecular biology. Watson and Crick, along with Maurice Wilkins, were awarded the Nobel Prize in 1962 for this work. (Sadly, Rosalind Franklin had died at the age of 37 in 1958 and was thus ineligible for the prize.) The beauty of the double helix model was that the structure of DNA suggested the basic mechanism of its replication.

➔ **Mastering Biology** HHMI Video: Great Discoveries in Science: The Double Helix



CONCEPT CHECK 16.1

1. Given a polynucleotide sequence such as GAATTC, explain what further information you would need in order to identify which is the 5' end. (See Figure 16.5.)
2. **VISUAL SKILLS** While trying to develop a vaccine for *S. pneumonia*, Griffith was surprised to discover the phenomenon of bacterial transformation. Based on the results in the second and third panels of Figure 16.2, what result was he expecting in the fourth panel? Explain.

For suggested answers, see Appendix A.

CONCEPT 16.2

Many proteins work together in DNA replication and repair

Hereditary information in DNA directs the development of your biochemical, anatomical, physiological, and, to some extent, behavioral traits. Your resemblance to your parents has its basis in the accurate replication of DNA prior to meiosis and therefore its transmission from your parents' generation to yours. Replication prior to mitosis ensures faithful transmission of genetic information from a parent cell to two daughter cells.

Of all nature's molecules, nucleic acids are unique in their ability to dictate their own replication from monomers. The relationship between structure and function is evident in the double helix: The specific complementary pairing of nitrogenous bases in DNA has a functional significance. Watson and Crick ended their classic paper with this statement: “It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.”¹ In this section, you will learn about the basic principle of **DNA replication**, the copying of DNA, as well as some important details of the process.

¹J. D. Watson and F. H. C. Crick, Molecular structure of nucleic acids: a structure for deoxyribose nucleic acids, *Nature* 171:737–738 (1953).